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Economic Load Dispatch

INTRODUCTION

The economic load dispatch problem involves the solution of two different problems. The first of these is the Unit Commitment or predispach problem wherein it is required to select optimally out of the available generating sources to operate, to meet the expected load and provide a specified margin of operating reserve over a specified period of time. The second aspect of economic dispatch is the on-line economic dispatch wherein it is required to distribute the load among the generating units actually paralleled with the system in such manner as to minimize the total cost of supplying the minute-to-minute requirements of the system.

It is this aspect of the economic dispatch problem that has been discussed in this chapter.

With large interconnection of the electric networks, the energy crisis in the world and continuous rise in prices, it is very essential to reduce the running charges of the electric energy *i.e.*, reduce the fuel consumption for meeting a particular load demand.

In the last chapter we made load flow study for the power system, in that for a particular load demand the generation at all the generator buses are fixed except at one generator bus known as slack, reference or swing bus where we allow the generation to take value within certain limits. In case of economic load dispatch the generations are not fixed but they are allowed to take values again within certain limits so as to meet a particular load demand with minimum fuel consumption. This means economic load dispatch problem is really the solution of a large number of load flow problems and choosing the one which is optimal in the sense that it needs minimum cost of generation. It is clear from this that since total cost of generation is a function of the individual generation of the sources which can take values within certain constraints, the cost of generation will depend upon the system constraint for a particular load demand. This means the cost of generation is not fixed for a particular load demand but depends upon the operating constraints of the sources. In fact the modern power system has to operate under various operational and network constraints. It is, therefore, best to understand the various constraints before actually taking up the economic load dispatching problem.

19.1 SYSTEM CONSTRAINTS

Broadly speaking there are two types of constraints: (i) Equality constraints, and (ii) Inequality constraints. Inequality constraints are of two types: (i) Hard type, and (ii) Soft type. The hard type are those which are definite and specific like the tapping range of an on-load tap changing transformer whereas soft type are those which have some flexibility associated with them like the nodal voltages and phase angles between the nodal voltages, etc. Soft inequality constraints have been very efficiently handled by penalty function methods.

Equality Constraints

The equality constraints are the basic load flow equations given by

$$P_p = \sum_{q=1}^n \{e_p(e_q G_{pq} + f_q B_{pq}) + f_p(f_q G_{pq} - e_q B_{pq})\}$$

$$Q_p = \sum_{q=1}^n \{f_p(e_q G_{pq} + f_q B_{pq}) - e_p(f_q G_{pq} - e_q B_{pq})\}$$

$$p = 1, 2, \dots, n$$

where e_p and f_p are the real and imaginary components of voltage at the p th node and G_{pq} and B_{pq} are the nodal conductance and susceptance between the p th and q th nodes.

Inequality Constraints

(a) *Generator Constraints:* The kVA loading on a generator is given by $\sqrt{P_p^2 + Q_p^2}$ and this should not exceed a prespecified value C_p because of the temperature rise conditions, i.e.,

$$P_p^2 + Q_p^2 \leq C_p^2$$

The maximum active power generation of a source is limited again by thermal consideration and also minimum power generation is limited by the flame instability of a boiler. If the power output of a generator for optimum operation of the system is less than a prespecified value $P_{\min}$ the unit is not put on the bus bar because it is not possible to generate that low value of power from that unit. Hence the generator powers P_p cannot be outside the range stated by the inequality, i.e.,

$$P_{p \min} \leq P_p \leq P_{p \max}$$

Similarly the maximum and minimum reactive power generation of a source are limited. The maximum reactive power is limited because of overheating of the rotor and minimum is limited because of the stability limit of the machine. Hence the generator reactive power Q_p cannot be outside the range stated by the inequality, i.e.,

$$Q_{p \min} \leq Q_p \leq Q_{p \max}$$

(b) *Voltage Constraints:* It is essential that the voltage magnitudes and phase angles at various nodes should vary within certain limits. The voltage magnitudes should vary within certain limits because otherwise most of the equipments connected to the system will not operate satisfactorily or additional use of voltage regulating devices will make the system uneconomical. Thus

$$|V_{p \min}| \leq |V_p| \leq |V_{p \max}|$$

$$\delta_{p \min} \leq \delta_p \leq \delta_{p \max}$$

where $|V_p|$ and δ_p stand for the voltage magnitude and phase angle at the p th node.

The normal operating angle of transmission line lies between 30° to 45° for transient stability reasons; therefore a higher limit is imposed on angle δ . A lower limit of δ assures proper utilization of transmission facility.

(c) *Running Spare Capacity Constraints*: These constraints are required to meet: (i) the forced outages of one or more alternators on the system, and (ii) the unexpected load on the system.

The total generation should be such that in addition to meeting load demand and losses a minimum spare capacity should be available *i.e.*,

$$G \geq P_p + P_{so}$$

where G is the total generation and P_{so} is some prespecified power. A well planned system is one in which this spare capacity P_{so} is minimum.

(d) *Transformer Tap Settings*: If an auto-transformer is used, the minimum tap setting could be zero and the maximum one, *i.e.*,

$$0 \leq t \leq 1.0$$

Similarly for a two winding transformer if tapings are provided on the secondary side,

$$0 \leq t \leq n$$

where n is the ratio of transformation. Phase shift limits of the phase shifting transformer

$$\theta_{p \min} \leq \theta_p \leq \theta_{p \max}$$

(e) *Transmission Line Constraints*: The flow of active and reactive power through the transmission line circuit is limited by the thermal capability of the circuit and is expressed as

$$C_p \leq C_{p \max}$$

where $C_{p \max}$ is the maximum loading capacity of the p th line.

(f) *Network Security Constraints*: If initially a system is operating satisfactorily and there is an outage, may be scheduled or forced one, it is natural that some of the constraints of the system will be violated. The complexity of these constraints (in terms of number of constraints) is increased when a large system is under study. In this case a study is to be made with outage of one branch at a time and then more than one branches at a time. The nature of constraints are same as voltage and transmission line constraints.

This completes more or less the study of various constraints. We now switch over to the actual problem of economic load dispatch. Since we are considering a system which is already in existence, the fixed costs like salaries and capital costs are neglected from the cost criterion. Also, we consider here only the thermal plants and, therefore, we are concerned only with the cost of fuel input to different generating stations.

This chapter will deal into the problem of optimization using the conventional B -coefficient method for loss evaluation.

In the past the practice was to load the machines such that the incremental cost of production for all the machines was same and thus transmission line was not considered an

integral part of the system. Later on, it was realized that a plant far from the load has to supply in addition to the load, transmission losses also and, therefore, this plant must be penalized. These transmission line losses are calculated by making use of the B -coefficients.

Before we take up the economic operation problem a few definitions are warranted. A simplified performance curve of a boiler turbine generator unit is given in Fig. 19.1. From the curve heat rate is defined as the ratio of fuel input to the corresponding power output and hence the units are million Btu per MWhr. From the heat rate curve another characteristic known as incremental fuel curve can be obtained. Incremental fuel rate is defined as

$$\text{Incremental fuel rate} = \frac{\Delta \text{input}}{\Delta \text{output}}$$

which means it is a ratio equal to a small change in input to the corresponding small change in output. Now as the incremental quantities tend to zero, incremental fuel rate tends to

$$\text{Incremental fuel rate} = \frac{d(\text{input})}{d(\text{output})} = \frac{dF}{dP}$$

where F is the fuel input in million Btu per hr and P is the power output in MW. The units associated with incremental fuel rate are, therefore, million Btu per MWhr and the incremental fuel cost is expressed in terms of Rs. per MWhr which is obtained by multiplying the incremental fuel rate by the fuel cost in Rs. per Btu.

Incremental production cost consists of the incremental fuel cost plus the incremental cost of labour, supplies, maintenance and water. Since it is difficult to express exactly these costs as a function of output and also since they form generally a small fraction of the incremental cost of fuel, the incremental cost of production will hitherto be considered equal to the incremental cost of fuel. The incremental efficiency is defined as the reciprocal of incremental fuel rate or incremental production cost and is given as

$$\text{Incremental efficiency} = \frac{(\text{output})}{(\text{input})} = \frac{dP}{dF}$$

Merit Order Scheduling

When this method is applied to economic scheduling of power plants, it assumes that the incremental cost of all the generators is constant over the full range or over successive discrete portions within the range. The economical way of meeting a load will be to load the machines in order of highest incremental efficiency. This method, therefore, needs forming of a table which could be looked into, for any load condition and does not need any complicated calculations. Normally the incremental cost curves are not constant; therefore, this method is not used and will not be discussed henceforth.

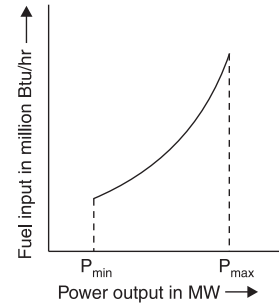


Fig. 19.1 Input-output curve.

19.2 ECONOMIC DISPATCH NEGLECTING LOSSES

The economic dispatch problem is defined as

$$\text{Min } F_T = \sum_{n=1}^n F_n \quad (19.1)$$

$$\text{Subject to } P_D = \sum_{n=1}^n P_n \quad (19.2)$$

where F_T is total fuel input to the system, F_n the fuel input to n th unit, P_D the total load demand and P_n the generation of n th unit.

By making use of Lagrangian multiplier the auxiliary function is obtained as

$$F = F_T + \lambda(P_D - \sum_{n=1}^n P_n)$$

where λ is the Lagrangian multiplier.

Differentiating F with respect to the generation P_n and equating to zero gives the condition for optimal operation of the system.

$$\begin{aligned} \frac{\partial F}{\partial P_n} &= \frac{\partial F_T}{\partial P_n} + \lambda(0 - 1) = 0 \\ &= \frac{\partial F_T}{\partial P_n} - \lambda = 0 \end{aligned}$$

$$\text{Since } F_T = F_1 + F_2 + \dots + F_n,$$

$$\therefore \frac{\partial F_T}{\partial P_n} = \frac{dF_n}{dP_n} = \lambda$$

and therefore the condition for optimum operation is

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2} = \dots = \frac{dF_n}{dP_n} = \lambda \quad (19.3)$$

Here $\frac{dF_n}{dP_n}$ = incremental production cost of plant n in Rs. per MWhr.

The incremental production cost of a given plant over a limited range is represented by

$$\frac{dF_n}{dP_n} = F_{nn}P_n + f_n$$

where F_{nn} = slope of incremental production cost curve, and

f_n = intercept of incremental production cost curve.

The equation (19.3) mean that the machines be so loaded that the incremental cost of production of each machine is same. It is to be noted here that the active power generation constraints are taken into account while solving the equations which are derived above. If these constraints are violated for any generator it is tied to the corresponding limit and the rest of the load is distributed to the remaining generator units according to the equal

incremental cost of production. The simultaneous solution of equations (19.2) and (19.3) gives the economic operating schedule. Therefore, any good technique for solving a set of linear equations can be used but since the inequality constraints have also to be taken into account, the following iterative method is used:

1. Assume a suitable value of λ^0 . This value should be more than the largest intercept of the incremental cost characteristics of the various generators.

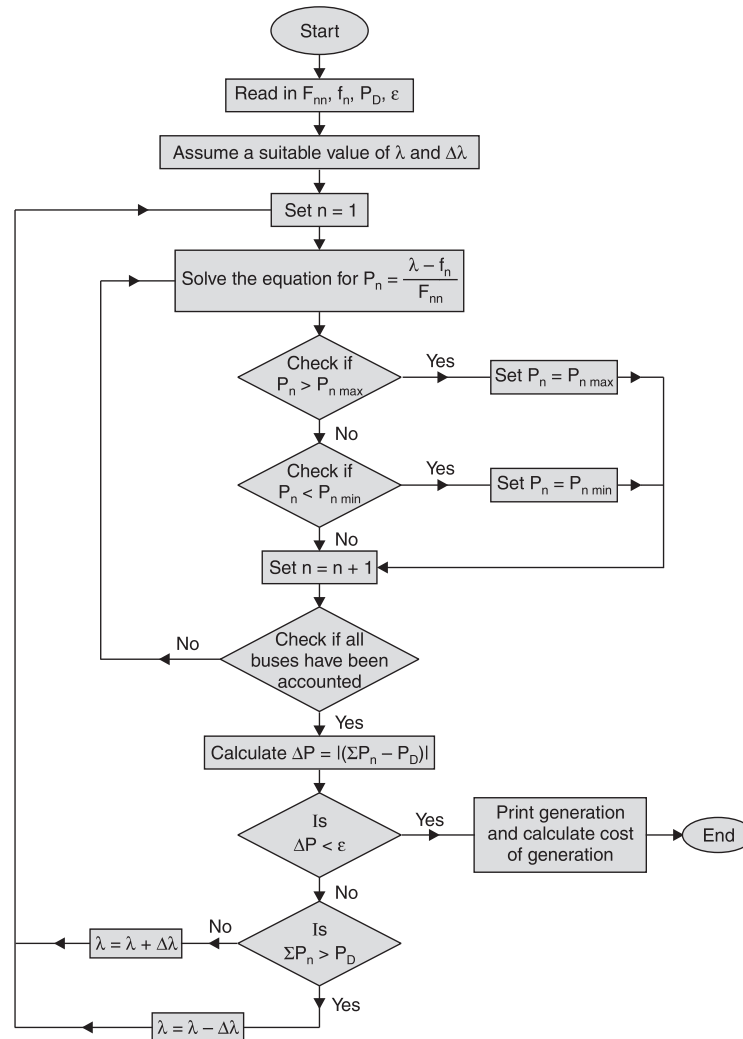


Fig. 19.2 Flow chart for economic scheduling: transmission loss neglected.

2. Compute the individual generations $P_1, P_2, \dots, P_n$ corresponding to incremental cost of production from equation (19.3). In case generation at any of the buses is violated, the generation of that generator is fixed at the limit violated during that iteration and the remaining load is distributed among the remaining generators.

3. Check if the equality

$$\sum_{n=1}^n P_n = P_D \text{ is satisfied.}$$

4. If not, make a second guess λ' and repeat the above steps. The selection of λ' in this step must of course be guided by the result in step (3). For example, if we find that the total generation is less than P_D then correct value of λ to be selected would be

$$\lambda_{\text{corr}} > \lambda^0$$

If equality is satisfied, the generations as calculated in step (2) give the optimum operating strategy. The flow chart is given in Fig. 19.2.

To illustrate the above procedure, the following problem is considered. There are two generators of 100 MW each with incremental characteristics:

$$\begin{aligned} \frac{dF_1}{dP_1} &= 2 + 0.012 P_1 \\ \frac{dF_2}{dP_2} &= 1.5 + 0.015 P_2 \end{aligned}$$

Minimum load on each unit is 10 MW, total load to be supplied is 150 MW. Determine the economic operating schedule.

Solution: Assume $\lambda = 2.8$ (more than the largest of the intercepts (2, 1.5). Determine generations P_1 and P_2 .

$$P_1 = \frac{2.8 - 2.0}{0.012} = 66.6 \text{ MW}$$

$$P_2 = \frac{2.8 - 1.5}{0.015} = 86.6 \text{ MW}$$

$$\therefore P_1 + P_2 = 66.6 + 86.6 = 153.2$$

Since generation is greater than the demand, therefore next approximation of λ should be less than 2.8. Also since load demand is 150 MW, the difference between P_D and P_n is not much. Therefore, value of λ should not be very much different from 2.8. Let it be 2.7.

With this, the generations are $58.4 + 80 = 138.4$ MW, which is less than the demand. Therefore, next approximation of λ should be greater than 2.7; say it is 2.75. Corresponding to this the generations are $62.5 + 83.4 = 145.9$ MW *i.e.*, the generations are smaller than the demand. Next approximation to $\lambda = 2.78$ which gives generations as $65 + 85.4 = 150.4$ MW. It is to be seen here that the generations are converging as better approximations to λ are chosen and finally a value between 2.77 and 2.78 will give the optimal generation schedule.

It can be seen that this method does not sense the location of changes in the loads. As long as the total load is fixed, irrespective of the location of loads, the solution will always be the same and in fact for this reason the solution may not be feasible in the sense that the load

voltages may not be within specified limits, the reactive power generations required also may not be within limits.

In case of an urban area where the load density is very high and the transmission distances are very small, the transmission loss could be neglected and the optimum strategy of generation could be based on the equal incremental production cost as outlined above. Whereas if the energy to be transported is over relatively larger distances with low load density, the transmission losses in some cases may amount 20 to 30% of the total load and it then becomes very essential to take these losses into account when developing an economic dispatch strategy.

19.3 OPTIMUM LOAD DISPATCH INCLUDING TRANSMISSION LOSSES

Before an optimum strategy for load scheduling is derived, the need for inclusion of losses is further stressed by the following example.

Consider Fig. 19.3 which consists of two identical generators *i.e.*, generators with identical incremental production cost. If generator 2 has a local load, according to equal incremental production criterion, the total load must be shared equally by both the generators, *i.e.*, each generator should supply half of the total load. The common sense tells us that it is more economical to let generator 2 supply most of the local load because generator 1 has to supply in addition to the load, the transmission losses also. Therefore, the criterion of sharing load by equal incremental production cost does not hold good under such situation and a strategy must be evolved which takes into account the transmission losses also.

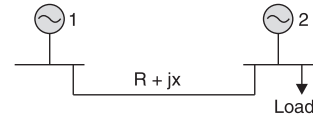


Fig. 19.3 Two identical generators connected through a transmission link.

The optimal load dispatch problem including transmission losses is defined as

$$\text{Min } F_T = \sum_{n=1}^n F_n \quad (19.4)$$

$$\text{Subject to } P_D + P_L - \sum_{n=1}^n P_n = 0 \quad (19.5)$$

where P_L is the total system loss which is assumed to be a function of generation and the other term have their usual significance.

Making use of the Lagrangian multiplier λ , the auxiliary function is given by

$$F = F_T + \lambda(P_D + P_L - \sum P_n)$$

The partial differential of this expression when equated to zero gives the condition for optimal load dispatch, *i.e.*,

$$\frac{\partial F}{\partial P_n} = \frac{\partial F_T}{\partial P_n} + \lambda \left(\frac{\partial P_L}{\partial P_n} - 1 \right) = 0$$

or

$$\frac{dF_n}{dP_n} + \lambda \frac{\partial P_L}{\partial P_n} = \lambda \quad (19.6)$$

Here the term $\frac{\partial P_L}{\partial P_n}$ is known as the incremental transmission loss at plant n and λ is known as the incremental cost of received power in Rs. per MWhr.

The equation (19.6) is a set of n equations with $(n + 1)$ unknowns. Here n generations are unknown and λ is also unknown. These equations are known as coordination equations because they coordinate the incremental transmission losses with the incremental cost of production.

To solve these equations the loss formula equation (19.7) is expressed in terms of generations and is approximately expressed as

$$P_L = \sum_m \sum_n P_m B_{mn} P_n \quad (19.7)$$

where P_m and P_n are the source loadings, B_{mn} the transmission loss coefficients. The formula is derived under the following assumptions:

1. The equivalent load current at any bus remains a constant complex fraction of the total equivalent load current.
2. The generator bus voltage magnitudes and angles are constant.
3. The power factor of each source is constant.

The solution of coordination equation (19.6) requires the calculation of $\partial P_L / \partial P_n$ which is obtained from equation (19.7) as

$$\frac{\partial P_L}{\partial P_n} = 2 \sum_m B_{mn} P_m \quad (19.8)$$

Also

$$\frac{dF_n}{dP_n} = F_{nn} P_n + f_n$$

$\therefore$ The coordination equations can be rewritten as

$$F_{nn} P_n + f_n + \lambda \sum_m 2B_{mn} P_m = \lambda \quad (19.9)$$

Collecting all coefficients of P_n , we obtain

$$P_n (F_{nn} + 2\lambda \sum_{m \neq n} B_{mn} P_m) = -\lambda (f_n + \lambda \sum_{m \neq n} 2B_{mn} P_m) - f_n + \lambda$$

Solving for P_n we obtain

$$P_n = \frac{1 - \frac{f_n}{\lambda} \sum_{m \neq n} 2B_{mn} P_m}{\frac{F_{nn}}{\lambda} + 2B_{nn}} \quad (19.10)$$

To arrive at an optimal load dispatching solution, the simultaneous solution of the coordination equations along with the equality constraint (19.5) should suffice and any standard matrix inversion subroutine could be used. But, because of the fact that plants might go beyond their loading conditions, it becomes necessary to solve a new set of equations and thus by the

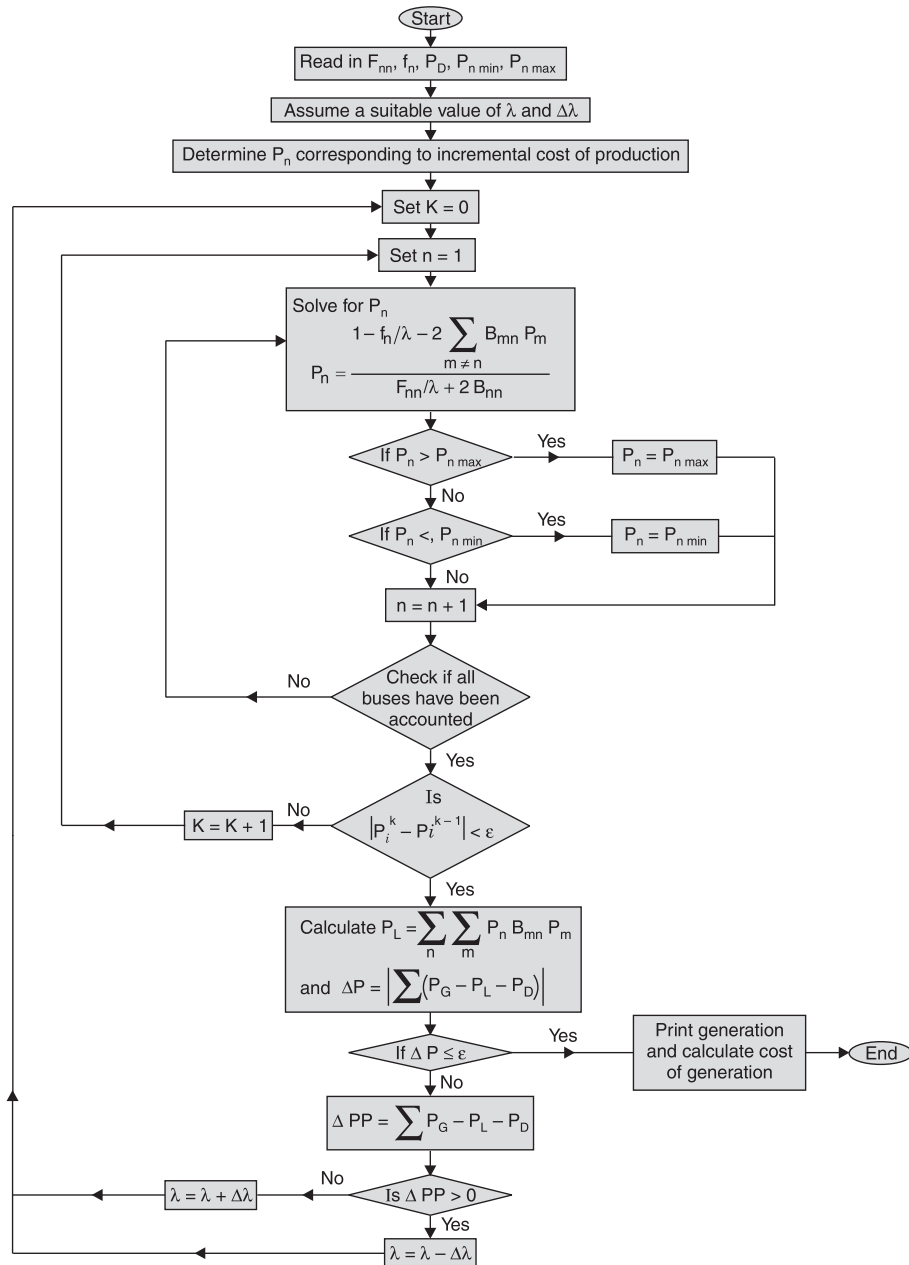


Fig. 19.4 Flow chart for the solution of coordination equations.

process of elimination this could be done. This would be very time consuming in a large interconnected system. Therefore, an iterative procedure would be used. The iterative procedure involves a method of successive approximation which rapidly converge to the correct solution. The following steps are required for the iterative procedure:

1. Assume a suitable value of λ^0 . This value should be more than the largest intercept of the incremental production cost of the various generators.
2. Calculate the generations based on equal incremental production cost.
3. Calculate the generation at all the buses using the equation

$$P_n = \frac{1 - \frac{f_n}{\lambda} - \sum_{m \neq n} 2B_{mn}P_m}{\frac{F_{nn}}{\lambda} + 2B_{nn}}$$

It is to be noted that the powers to be substituted on the right hand side during zeroth iteration correspond to the values as calculated in step 2. For subsequent iterations the values of powers to be substituted correspond to the powers as calculated in the previous iteration. In case any of the generations violates the limit the generation of that generator is fixed at the limit violated.

4. Check if the difference in power at all generator buses between two consecutive iterations is less than a prespecified value. If not, go back to step 3.
5. Calculate losses using the relation

$$P_L = \sum_m \sum_n P_n B_{mn} P_m$$

and calculate

$$\Delta P = | \Sigma P_G - P_L - P_D |$$

6. If ΔP is less than ϵ , stop calculation and calculate cost of generation with these values of powers.

7. Update value of λ and go back to step 3. The flow chart is given in Fig. 19.4.

The iterative procedure is illustrated for a simple two-plant system. Assume that the loss formula coefficients in 1/MW units are given by

m	n	B_{mn}
1	1	0.0015
1	2	- 0.0005
2	2	0.0025

Also assume

$$\frac{dF_1}{dP_1} = 0.01P_1 + 2.0$$

and

$$\frac{dF_2}{dP_2} = 0.01P_2 + 1.5$$

The objective is to determine the operating schedule corresponding to $\lambda = 2.6$ (say).

$$\begin{aligned}
 &\text{and} & P_1 + P_2 &= 50 \\
 &\text{i.e.,} & 0.048 P_1 + 8 &= 0.08 P_2 + 6 \\
 &\text{and} & P_1 + P_2 &= 50
 \end{aligned}$$

From these equations,

$$P_1 = 15.625 \text{ MW and } P_2 = 34.375 \text{ MW}$$

$$\therefore F_1 = 210.868 \text{ million Btu per hr.}$$

$$\text{and } F_2 = 373.5 \text{ million Btu per hr.}$$

(ii) When load is 150 MW, the equations are

$$0.048 P_1 + 8 = 0.08 P_2 + 6$$

$$\text{and } P_1 + P_2 = 150$$

$$\therefore P_1 = 78.126 \text{ MW and } P_2 = 71.874 \text{ MW}$$

$$\text{and } F_1 = 851.496 \text{ million Btu/hr.}$$

$$F_2 = 757.87 \text{ million Btu/hr.}$$

$$\begin{aligned}
 \therefore \text{Total cost} &= \text{Rs. } (210.868 + 373.5 + 851.496 + 757.87) \times 12 \times 2 \\
 &= \text{Rs. } 52649.61 \quad \text{Ans.}
 \end{aligned}$$

Physical Interpretation of Coordination Equations

The physical interpretation of the coordination equations can be understood with the help of Fig. 19.5. There are n number of plants connected to a hypothetical load through a transmission network. The incremental cost of production of n th plant at the busbar is dF_n/dP_n . The plant n incurs an incremental transmission loss of $\partial P_L/\partial P_n$ in supplying the next increment of load. Let ΔP_D be the change in load and if this change is supplied by plant n alone and say in doing so the generation at plant n required is ΔP_n . This ΔP_n includes the transmission losses in addition to the increase in demand by ΔP_D . The cost of power at the plant bus will be

$$\frac{dF_n}{dP_n} \Delta P_n = \text{Rs./hr}$$

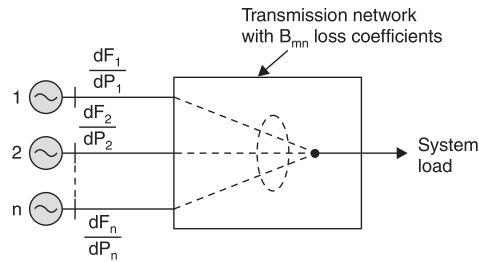


Fig. 19.5 Physical interpretation of coordination equations.

But since power received is only ΔP_D , therefore the cost of received power will be

$$\lambda = \frac{dF_n}{dP_n} \cdot \Delta P_n \cdot \frac{1}{\Delta P_D} = \frac{dF_n}{dP_n} \cdot \frac{\Delta P_n}{\Delta P_D} \text{ Rs./MWhr}$$

Since $\Delta P_D = \Delta P_n - \Delta P_L$

$$\therefore \lambda = \frac{dF_n}{dP_n} \frac{\Delta P_n}{\Delta P_n - \Delta P_L} = \frac{dF_n}{dP_n} \frac{1}{1 - \frac{\Delta P_L}{\Delta P_n}}$$

and in the limit when $\Delta P_n \rightarrow 0$

$$\lambda = \frac{dF_n}{dP_n} \frac{1}{1 - \frac{\partial P_L}{\partial P_n}} = \frac{dF_n}{dP_n} \cdot L_n$$

which is same as the coordination equation and from this treatment penalty factor for plant n can be defined as the ratio of the small change in power at plant n to the small change in received power when generation at plant n alone is changed for meeting this change in load.

When transmission losses are included and coordinated, the following points must be kept in mind for economic load dispatch solution:

1. Whereas the incremental cost of production of a plant is always positive, the incremental transmission losses can be both positive or negative.
2. The individual generators will operate at different incremental costs of production.
3. The generation with highest positive incremental transmission loss will operate at the lowest incremental cost of production.

The B_{mn} coefficients are the loss coefficients and for an n generator system the coefficient is an $n \times n$ symmetric matrix

$$B_{mn} = \begin{bmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & B_{22} & \cdots & B_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n1} & B_{n2} & \cdots & B_{nn} \end{bmatrix}$$

The diagonal elements are all positive and strong as compared with the off diagonal elements which mostly are negative and are relatively weaker. These coefficients are determined for a large system by an elaborate computer programme starting from the assembly of the open circuit impedance matrix of the transmission network which is quite lengthy and time consuming and is beyond the scope of this book. Besides, the formulations of B -coefficients are based on several assumptions and do not take into account the actual conditions of the system, the solution for the plant generations cannot be expected to be the best for minimum cost of generation.

B -coefficients have been developed by applying tensors to power system wherein the interconnected system is reduced to one with sources equal to the actual number of sources but loads equal to one equivalent hypothetical load.

We develop here simple equations for loss coefficients based on several assumptions for systems having any number of loads and sources. We will develop, to begin with, the equations for two generating stations and an indefinite number of loads and finally will generalize for any number of generators and loads. Consider Fig. 19.6 which indicates two generating plants connected to a transmission network which is terminated by an arbitrary number of loads.

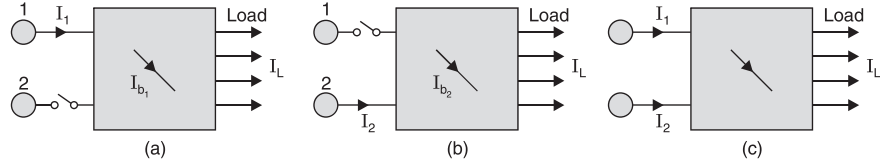


Fig. 19.6 Schematic diagram of a network with two generators and large number of loads. One branch of the network is shown.

Consider one three phase line b . The total load current I_L is supplied by source 1 and the current in line b is I_{b_1}

$$\text{Let} \quad \alpha_{b_1} = \frac{I_{b_1}}{I_L} \quad (19.13)$$

Similarly when source 2 alone supplies load current I_L and if I_{b_2} is the current through the same line b , then

$$\alpha_{b_2} = \frac{I_{b_2}}{I_L} \quad (19.14)$$

When both the plants are connected to supply the load current, the current through the branch b , using the principle of superposition, will be

$$I_b = \alpha_{b_1} I_1 + \alpha_{b_2} I_2 \quad (19.15)$$

where I_1 and I_2 are the currents from plant 1 and 2 respectively and α_{b_1} and α_{b_2} are the current distribution factors.

Here we make two simplifying assumptions for deriving the loss coefficients.

(i) The ratio X/R for all the transmission lines is the same.

(ii) The phase angle of all the load currents is the same.

The net effect of these two assumptions is that the load currents and branch currents are in phase and hence the current distribution factors are real.

$$\text{Let} \quad I_1 = |I_1| \cos \theta_1 + j |I_1| \sin \theta_1$$

$$\text{and} \quad I_2 = |I_2| \cos \theta_2 + j |I_2| \sin \theta_2 \quad (19.16)$$

where θ_1 and θ_2 are the phase angles of currents I_1 and I_2 with reference to a common phasor.

$$\begin{aligned} \therefore I_b &= \alpha_{b_1} I_1 + \alpha_{b_2} I_2 \\ &= \alpha_{b_1} |I_1| \cos \theta_1 + \alpha_{b_2} |I_2| \cos \theta_2 + j [\alpha_{b_1} |I_1| \sin \theta_1 + \alpha_{b_2} |I_2| \sin \theta_2] \end{aligned}$$

$$\text{or} \quad I_b^2 = \alpha_{b_1}^2 |I_1|^2 + \alpha_{b_2}^2 |I_2|^2 + 2\alpha_{b_1} \alpha_{b_2} |I_1| |I_2| \cos (\theta_1 - \theta_2) \quad (19.17)$$

Since $I_1 = \frac{P_1}{\sqrt{3} |V_1| \cos \phi_1}$ and $I_2 = \frac{P_2}{\sqrt{3} |V_2| \cos \phi_2}$ and if R_b is the resistance of branch b , the total loss will be

$$P_L = \sum_{b=1}^b 3 I_b^2 R_b$$

where Σ is the summation of losses in all the branches and ϕ_1 and ϕ_2 are the p.f. angles of plants 1 and 2 respectively.

$$P_L' = \sum \alpha_{b_1}^2 \frac{P_1^2 R_b}{|V_1|^2 \cos^2 \phi_1} + \sum \alpha_{b_2}^2 \frac{P_2^2 R_b}{|V_2|^2 \cos^2 \phi_2} + \sum \frac{2P_1 P_2 R_b \alpha_{b_1} \alpha_{b_2} \cos(\theta_1 - \theta_2)}{|V_1| |V_2| \cos \phi_1 \cos \phi_2} \quad (19.18)$$

$$= B_{11} P_1^2 + B_{22} P_2^2 + 2B_{12} P_1 P_2$$

where

$$B_{11} = \sum \frac{\alpha_{b_1}^2 R_b}{|V_1|^2 \cos^2 \phi_1} = \frac{1}{|V_1|^2 \cos^2 \phi_1} \sum_b \alpha_{b_1}^2 R_b$$

$$B_{22} = \frac{1}{|V_2|^2 \cos^2 \phi_2} \sum_b \alpha_{b_2}^2 R_b$$

and

$$B_{12} = \frac{1}{|V_1| |V_2| \cos \phi_1 \cos \phi_2} \sum_b \alpha_{b_1} \alpha_{b_2} R_b$$

Therefore, if there are n number of sources the general loss coefficients will be

$$B_{nn} = \frac{1}{|V_n|^2 \cos^2 \phi_n} \sum_b \alpha_{b_n}^2 R_b \quad (19.19)$$

and

$$B_{mn} = \frac{1}{|V_m| |V_n| \cos \phi_m \cos \phi_n} \sum_b \alpha_{b_m} \alpha_{b_n} R_b \quad (19.20)$$

Whenever there are wide variations in the operating conditions, the various assumptions for evaluation of B -coefficients may cause errors in loss calculation, it is desirable to obtain one or two additional sets of loss coefficients for such conditions.

Many power companies, however, obtain reasonably accurate results assuming one set of coefficients corresponding to a typical operating condition.

19.4 EXACT TRANSMISSION LOSS FORMULA

Dopezo et al. have derived an exact formula for calculating transmission losses by making use of the bus powers and the system parameters. Let S_i be the total injected bus power at bus i and is equal to the generated power minus the load at bus i . The summation of all such powers over all the buses gives the total losses of the system, i.e.,

$$P_L + jQ_L = \sum_{i=1}^n S_i = \sum_{i=1}^n V_i I_i^* = V_{\text{bus}}^T I_{\text{bus}}^* \quad (19.21)$$

Here P_L and Q_L are the real and reactive power loss of the system. V_{bus} and I_{bus} are the column vectors of voltages and currents of all the buses.

Now

$$V_{\text{bus}} = Z_{\text{bus}} I_{\text{bus}} \quad (19.22)$$

where Z_{bus} is the bus impedance matrix of the transmission network and is given by

$$Z_{\text{bus}} = R + jX = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{nn} \end{bmatrix} + j \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix}$$

From equations (19.21) and (19.22), we have

$$\begin{aligned} P_L + jQ_L &= I_{\text{bus}}^T Z_{\text{bus}}^T I_{\text{bus}}^* \\ &= I_{\text{bus}}^T Z_{\text{bus}} I_{\text{bus}}^* \end{aligned} \quad (19.23)$$

The last step can be written because Z_{bus} is a symmetric matrix and, therefore,

$$Z_{\text{bus}}^T = Z_{\text{bus}}.$$

The bus current vector I_{bus} can also be written as the sum of a real and reactive component of current vectors, *i.e.*,

$$I_{\text{bus}} = I_p + jI_q = \begin{bmatrix} I_{p1} \\ I_{p2} \\ \vdots \\ I_{pn} \end{bmatrix} + j \begin{bmatrix} I_{q1} \\ I_{q2} \\ \vdots \\ I_{qn} \end{bmatrix}$$

The equation for the loss can be written as

$$P_L + jQ_L = (I_p + jI_q)^T (R + jX) (I_p - jI_q)$$

Separating out the real part P_L from the above matrix product, we get

$$P_L = I_p^T R I_p + I_p^T X I_q + I_q^T R I_q - I_q^T X I_p$$

Since X is a symmetric matrix,

$$I_p^T X I_q = I_q^T X I_p$$

Hence

$$P_L = I_p^T R I_p + I_q^T R I_q \quad (19.24)$$

This expression (19.24) can be rewritten by using index notation as

$$P_L = \sum_{i=1}^n \sum_{k=1}^n r_{jk} (I_{pj} I_{pk} + I_{qj} I_{qk}) \quad (19.25)$$

The transmission loss has been expressed in terms of bus currents. In an actual power plant, the system operators usually know the bus powers and the nodal voltages; therefore, it is more practical to express P_L in terms of these quantities.

For bus powers at bus i we have

$$P_i + jQ_i = V_i I_i^* = V_i (I_{pi} - jI_{qi}) = |V_i| (\cos \delta_i + j \sin \delta_i) (I_{pi} - jI_{qi})$$

where δ_i is the phase angle of voltage V_i with respect to the reference voltage *i.e.*, the slack bus voltage. Now separating the real and imaginary parts, we have

$$P_i = |V_i| \cos \delta_i I_{pi} + |V_i| I_{qi} \sin \delta_i$$

and

$$Q_i = |V_i| I_{pi} \sin \delta_i - |V_i| I_{qi} \cos \delta_i$$

Solving for I_{pi} and I_{qi} , we get

$$I_{pi} = \frac{1}{|V_i|} (P_i \cos \delta_i + Q_i \sin \delta_i)$$

$$I_{qi} = \frac{1}{|V_i|} (P_i \sin \delta_i - Q_i \cos \delta_i)$$

We have expressed here the real and imaginary components of bus currents in terms of bus powers and the bus voltages. Substituting these values in the expression for power loss, we get after some algebraic manipulations,

$$P_L = \sum_{\substack{j=1 \\ k=1}}^n \alpha_{jk} (P_j P_k + Q_j Q_k) + \beta_{jk} (Q_j P_k - P_j Q_k) \quad (19.26)$$

where

$$\alpha_{jk} = \frac{r_{jk}}{|V_j| |V_k|} \cos (\delta_j - \delta_k)$$

and

$$\beta_{jk} = \frac{r_{jk}}{|V_j| |V_k|} \sin (\delta_j - \delta_k)$$

By making use of this expression for transmission losses along with the coordination equations, the problem of optimal load dispatch can be solved.

It can be seen that even though the formulation for the transmission loss is exact, the method requires the calculation of bus impedance matrix which is time consuming and needs more computer memory.

19.5 MODIFIED COORDINATION EQUATIONS

Another technique for solving the optimal load dispatch problem as suggested by Kirchmayer is through the use of modified coordination equations. These equations are derived as follows:

Transmission loss is the algebraic sum of the powers at all the nodes *i.e.*, it is function of powers at all the nodes and therefore

$$P_L = \sum_{i=1}^n P_i \quad (19.27)$$

Here P_i is positive for generator bus and negative for load bus.

$$\therefore dP_L = \sum_{i=1}^n \frac{\partial P_L}{\partial P_i} dP_i \quad (19.28)$$

Say, in an interconnected system the bus powers of only two plants P_j and P_n can be changed by small amounts keeping the powers at all other buses fixed then,

$$dP_{L,j,n} = \frac{\partial P_L}{\partial P_j} dP_j + \frac{\partial P_L}{\partial P_n} dP_n \quad (19.29)$$

The incremental loss $dP_{L,j,n}/dP_j$ is defined as the ratio of change in loss to the change in generation at plant j when power is transferred from plant j to plant n , with the generations of other plants and load keeping fixed. It is clear that

$$dP_{L,j,n} = dP_j + dP_n \quad (19.30)$$

Using equations (19.29) and (19.30), we get

$$dP_j + dP_n = \frac{\partial P_L}{\partial P_j} dP_j + \frac{\partial P_L}{\partial P_n} dP_n$$

or

$$dP_j \left(1 - \frac{\partial P_L}{\partial P_j} \right) + dP_n \left(1 - \frac{\partial P_L}{\partial P_n} \right) = 0$$

or
$$\frac{dP_j}{dP_n} = - \frac{1 - \partial P_L / \partial P_n}{1 - \partial P_L / \partial P_j} \quad (19.31)$$

Also since from the coordination equations,

$$\begin{aligned} \frac{dF_j}{dP_j} \cdot \frac{1}{\left(1 - \frac{\partial P_L}{\partial P_j}\right)} &= \lambda = \frac{dF_n}{dP_n} \cdot \frac{1}{1 - \frac{\partial P_L}{\partial P_n}} \\ \frac{\frac{dF_n}{dP_n}}{\frac{dF_j}{dP_j}} &= \frac{1 - \frac{\partial P_L}{\partial P_n}}{1 - \frac{\partial P_L}{\partial P_j}} = - \frac{dP_j}{dP_n} = - \frac{dP_j}{dP_{Lj,n} - dP_j} \\ &= \frac{1}{1 - \frac{dP_{Lj,n}}{dP_j}} \end{aligned}$$

or
$$\frac{dF_n}{dP_n} = \frac{1}{1 - \frac{dP_{Lj,n}}{dP_j}} \cdot \frac{dF_j}{dP_j} = \mu \quad (19.32)$$

Here plant n is taken as the reference plant.

From the above expression it is clear that for economic load dispatch the condition required is that the incremental cost of power at plant bus n is equal to the incremental cost of power at plant bus j corrected for the effect of the incremental transmission loss involved in swinging generation between plants j and n . These equations are known as modified coordination equations. The modified coordination equation can be rewritten as

$$\frac{dF_n}{dP_n} = - \frac{dF_j}{dP_j} \cdot \frac{\Delta P_j}{\Delta P_n} = \mu$$

The solution of modified coordination equation requires the calculation of $\Delta P_j / \Delta P_n$ terms. The solution of these equations is explained as follows: The flow chart is given in Fig. 19.7.

1. Given a demand, assuming no transmission losses, the generations are determined according to equal incremental production cost, say $P_1, P_2, \dots, P_n$ are the generations of n plants. Thus solving the equation

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2} = \dots = \frac{dF_n}{dP_n}$$

and

$$\sum_{n=1}^n P_n = P_D$$

where P_D is the given system load demand. The generation $P_1, P_2, \dots, P_n$ are determined. Choosing n th plant as the reference plant, the incremental cost μ at the reference plant is given by

$$\mu = \frac{dF_n}{dP_n} = F_{nn} P_n + f_n$$

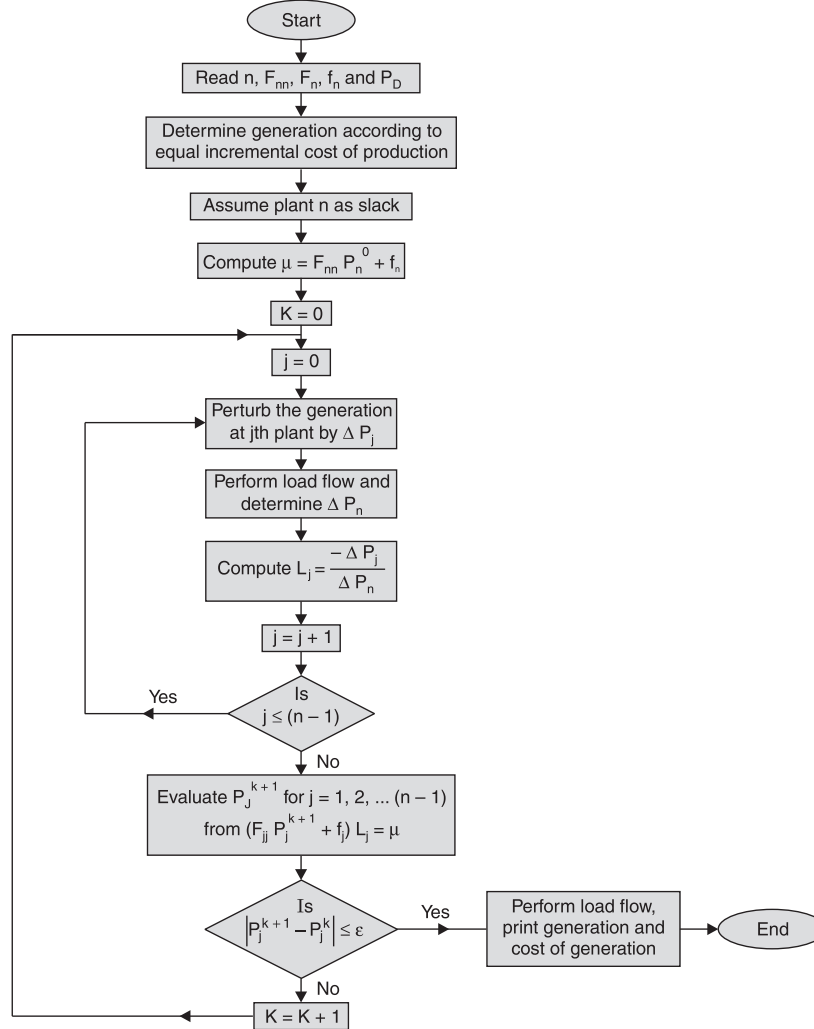


Fig. 19.7 Flow chart for the solution of modified coordination equations.

2. Taking the n th plant bus as the reference or slack bus, a load flow using Newton-Raphson technique is performed with specified generations as obtained in step 1. Let the slack bus power be P_n' .

3. Increase the generation at bus 1 by a small amount ΔP_1 and keeping all other generations constant as per step 1 and taking plant n as the slack bus perform the load flow again. Say the slack bus power be P_n'' .

The procedure is repeated by swinging generation between other plants and the reference plant n , one at a time, for obtaining the quantities $-\frac{\Delta P_j}{\Delta P_n}$.

4. The equations $\frac{dF_n}{dP_n} = -\frac{\Delta P_j}{\Delta P_n} \cdot \frac{dF_j}{dP_j}$ are solved. The solution gives new generation $P_1, P_2, \dots, P_j$. The first iteration for generation schedule is complete.

5. With the reference bus remaining unchanged and with the value of μ remaining constant as chosen in step 1, the procedure from 1 to 4 is repeated until the generation of two consecutive iterations lie within a certain specified tolerance, *i.e.*, mathematically,

$$|P_j^{k+1} - P_j^k| \leq \varepsilon$$

where K is the iteration count and ε is the prespecified tolerance. When convergence is reached the next step is followed, otherwise steps 2 to 4 are repeated.

6. With P_j^{k+1} as generation, the final load flow solution is obtained and hence the cost of generation is evaluated.

19.6 AUTOMATIC LOAD DISPATCHING

It has already been explained that economic load dispatching is that aspect of power system operation wherein it is required to distribute the load among the generating units actually paralleled with the system in such a manner as to minimise the cost of supplying the minute-to-minute requirements of the system. In a large interconnected system it is humanly impossible to calculate and adjust such generations and hence the help of digital computer system along with analogue devices is sought and the whole process is carried out automatically; hence the term automatic load dispatching is used for the purpose.

An energy system is normally controlled by a central control centre which coordinates the operations of several area centres. The objective is to minimise the cost of supplying electricity to the load points while ensuring security of supply against loss of generation and transmission capacity and also maintaining the voltage and frequency of the system within specified limits. Since the interconnection is growing bigger and bigger in size with time, the control engineer has to make adjustments to various parameters in the system. Hence it has become imperative to make use of automatic control for load dispatch problem.

The chosen control system is invariably based on a digital computer working on-line. The presence of the control engineer is required only to enter certain data. The rest of the control calculations can proceed and the generation instructions calculated, transmitted and implemented at the power stations automatically.

The block diagram of an automatic control system used for load dispatching is shown in Fig. 19.8.

Computer: The computer predicts the load and suggests economic loading. It transmits information to machine controllers.

Data Input: The computer receives a lot of essential data from the telemetering system and from the paper tape. Telemetering data comes to the computer either as analogue signals representing line power flows, plant outputs or as signal bits indicating switch or isolator positions. The system is entirely automatic. Paper tape stores all the basic data required, *e.g.*, the system parameters, load prediction, security constraints, etc. and finally asking the stations to switch the machine controllers to automatic control.

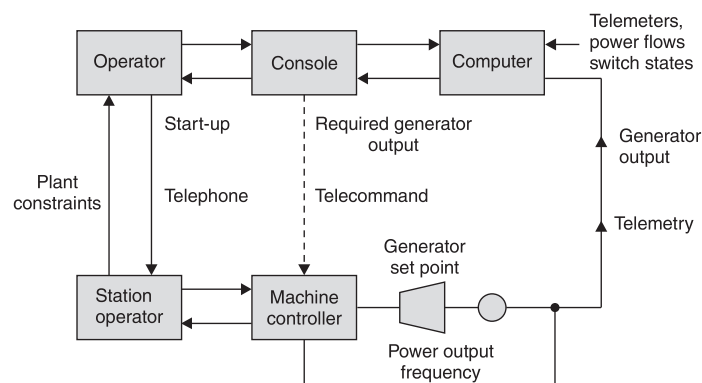


Fig. 19.8 Schematic diagram of automatic load dispatching components.

Console: The console is the component through which the operator can converse with the computer. He can obtain certain information required for some action to be taken under emergency condition should such a condition arise in the system, or he can put data into it if needed. The instructions being issued and the state of the energy system (tie-line loadings, plant outputs, nodal voltages, etc.) can be examined on cathode ray displays associated with the console.

The console has the facilities of security checking and load flows for the network calculations. Such a facility can be called manually at almost any instant of time and the results may be recorded automatically after a specified interval of time, if required. The security check (loss of generator or loss of one or more than one transmission links) reveals and displays that if one or more than one particular branches suffer outage, at least one of the remaining branches will be overloaded.

Machine Controllers: The computer sends instructions regarding the optimal generation to the machine controller at regular intervals which in turn implements them. Control on each machine is applied by a closed loop system which uses a measure of actual power generated and which operates through a conventional speeder motor. These are referred to as controller power loops.

In the power frequency loop an error signal proportional to the difference between the derived and actual frequency and power is developed. A summed error signal is formed from these two components and is converted in the motor controller to a train of pulses that are applied to a speed governor reference setting motor called the speeder motor. The duration and amplitude of these pulses are fixed but the pulse rate is made proportional to the summed

error signal. The pulses are applied as 'raise' or 'lower' command to the speeder motor in accordance with the error signal and thus the output of the generators is increased or decreased accordingly.

19.7 POWER LINE CARRIER COMMUNICATION (PLCC)

The term power line carrier is used to represent the entire process of communication which uses high voltage overhead power lines as the means of transmission. The power lines offer the following advantages:

- (i) Lines have thicker cross section of wire and hence attenuation of signals is not much.
- (ii) Leakage is negligible even under wet weather conditions as lines are insulated with high voltage insulators.
- (iii) Lines are disrupted during foul weather conditions as conductors are strung on very robust tower structures.
- (iv) As the phases are separated from each other quite appreciably, cross-talks between lines is practically avoided.
- (v) The cost of providing extra lines for the purpose is avoided.

However, there are certain difficulties associated with the transmission of communication signal over the power lines:

- (i) Since the power is being transmitted at relatively high voltages, the operation on these lines may prove to be dangerous to human lives and also to telephone apparatus.
- (ii) During transient operation of system *e.g.*, switching transients, surge voltages or corona phenomenon etc. the presence of higher harmonics in power currents may interfere with the communication signal.

These difficulties, of course, have been overcome by selecting suitable coupling capacitors and carrier frequencies.

The power line carrier communication finds application in telemetering, power line protection, telecontrol etc. The application of PLCC to power line protection has already been discussed in Chapter 14.

The power lines may have higher harmonics due to switching surges and corona loss on overhead lines. These frequencies generally lie between 100 Hz to 50 kHz and, therefore, if carrier frequencies are chosen in this region, noise introduced in carrier signal would be very large and hence carrier frequency in the range 30 kHz to 500 kHz are used. In order that these carrier signals do not interfere into the adjoining section of lines and also that these carrier signals are not lost by being shorted by low impedance of transformer or generator at the end of the line, line traps are used which offer very high impedance to carrier frequency and low impedance to power frequency. Similarly the coupling capacitors are so designed that they offer very high impedance to power frequency but low to carrier frequencies. The two most commonly employed coupling methods are centre-phase to ground and adjacent-phase to phase as shown in Fig. 19.9.

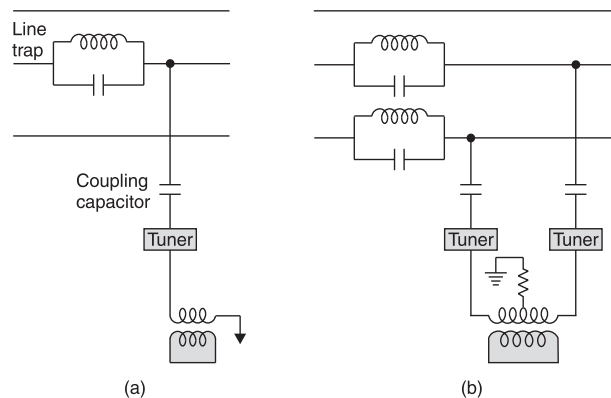


Fig. 19.9 Coupling circuits (a) centre-phase to ground and (b) adjacent phase to phase.

For centre-phase to ground only one phase line is required and ground is used as the return path whereas for adjacent phase to phase connection two phase lines are used for the go and return paths. Thus only half of the coupling components are required in the former method as compared with the latter one. Also with centre phase to ground method three separate carrier sets can be connected with three phase lines. But the adjacent phase to phase method has got less attenuation and transmission characteristics are fairly constant, radiation loss is smaller and signal to noise ratio (S/N) is high. For these reasons, even though adjacent phase to phase method is costlier but is normally used.

For efficient coupling at carrier frequencies, it is necessary to cancel the reactance of the coupling capacitor by means of inductance in the line matching unit and to match the impedance of the power line. Power line impedances are of the order of 300 to 500 ohms (phase to ground and phase to phase) respectively. The transformer included in the line matching unit are provided with several taps to facilitate optimum impedance matching between the terminal and the power line. Proper impedance match at both parts of the tuning unit is important not only to assure maximum power transfer but also to prevent undesirable standing waves due to reflections at a mismatched junction. The line matching unit is normally located close to the coupling capacitor in the switchyard, and it must be weather proof. For phase to phase coupling, it is common to employ two line matching units.

Different Methods of Carrier Transmission Over Power Lines

There are two methods of transmitting carrier signals over power lines between different stations in the power grid:

- (i) Fixed Frequency System; (ii) Wave Change-Over System.

In the first system, transmission of signals in different sections are carried out by different carrier frequencies as shown in Fig. 19.10.

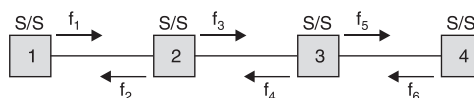


Fig. 19.10 Fixed frequency system.

Transmission between S/S 1 and 2 is carried out by carrier frequencies f_1, f_2 that between S/S 2 and 3 f_3, f_4 and between S/S 3 and 4, f_5 and f_6 and so on. In this method signals are sent between two consecutive stations by the allotted carrier frequency and the carrier is demodulated at the second station and the signal is used to modulate again the carrier frequency allotted for the second section and is retransmitted on this carrier and so on. The fixed frequency system has an advantage in that it is possible to have simultaneous transmission of signals in different sections and quality of reception is better as signals are amplified at each station. But it is a costlier system as two carrier sets are required at each intermediate substation.

In the second system (Fig. 19.11), two carrier frequencies f_1 and f_2 are used for transmission in two directions between any two stations. It is so planned that the carrier oscillator of frequency f_1 is used for sending the signal from the transmitter and the carrier oscillator of frequency f_2 is used for demodulation purpose to get back the signal in the receiver in the case of the calling station. As soon as the calling station picks up the telephone set for calling, the transmitter and the receiver side of the calling station are connected to these carrier oscillators. When the called station receives ring, its transmitter is provided with the carrier oscillator of frequency f_2 and its receiver is provided with the carrier oscillators of frequency f_1 .

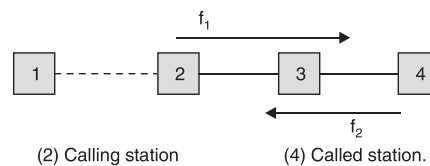


Fig. 19.11 Wave change over system.

In this way, transmission in both the directions are carried out by two carrier frequencies in all sections between any two substations.

The ground wire for protection of power lines against lightning stroke can also be used as a transmission medium by insulating the conductors from the tower with insulators having breakdown (equivalent air gap) of 15 kV to 25 kV. With this, the main function of the ground wire as a shield wire is not sacrificed. A benefit from using overhead ground wire for communication is the reduction of power frequency drainage currents induced in them due to the power lines.

The ground wire when insulated results in a slight increase in zero-sequence impedance of the power line which may cause an increase in over voltages when a short circuit involving ground occurs on the power line.

A suitable coupling device must be used at each ground wire terminal to provide a drainage path to ground for the induced power frequency influence and for lightning currents while coupling the carrier signal to the shield wires. The cost of coupling equipments in case of ground wire is much smaller and is a constant irrespective of the operating voltage. Another advantage of ground wire as a medium of transmission is that it does not require a power line outage when maintenance is required whereas this is always required for maintenance of phase wire equipment.

Whenever two ground wires are used, it is desirable to transpose the wires after every 10 to 30 km. (depending upon the experience about a particular system). With this the interference of power lines with the ground wires used as communication channels will be eliminated.

The insulated ground wires are more prone to foul weather conditions in regard to attenuation of signals as compared to the phase wires. These wires have been used for supervising control, telemetry and alarm functions, but there is limited experience in regard to its application for protective relay channels.

Bundled conductors may also be used as transmission channel for communications. Conductors of the same phase insulated from each other would be used and then ground will not be used as the return path. This scheme is being operated in USSR. The technique offers the following advantages:

- (i) Less interference with or from radio services in the PLC band.
- (ii) Lower fair weather lines loss.
- (iii) Heavy reduction in cross talk.

It is hoped that bundled conductors may prove to be an important technique in future application of power line carrier particularly in ice free areas.

Modern power system has grown quite complex. The awareness of power system engineer for a good quality of reliable supply has forced him to adopt some automatic means so that he not only operates his system reliably but most economically. For this he has to know the state of the system so that he can regulate the system in that direction. The state (frequency, voltage, active power, reactive power flows, position of C.B., switches etc.) of such a complex system can be obtained by telemetering only. No manual method can handle such a complex job.

For the sake of convenience, the whole system is divided into a number of areas or groups each area having an area (or group) control station and finally these stations are centrally controlled (or coordinated) by what are known as master controlled stations. These different control stations are located at suitable points in the network and they control such items as voltage, power, frequency on getting information from distant points regarding condition of the system by telemetering.

There are some informations which are to be telemetered to group control stations only; others are to be telemetered to master control station only; yet there are other informations to be telemetered to both the stations.

Informations telemetered to group control station:

- (i) Generated active and reactive power of each station in the group.
- (ii) Net active and reactive power transfer from the group.
- (iii) Active and reactive power flow through the tie lines, alongwith direction of flow and an overload alarm.

Information to be telemetered to control station:

- (i) Voltage and frequency from selected points.
- (ii) Total generated active and reactive power of the system.
- (iii) Tie-line loadings (active and reactive powers) from selected points.

Informations to be telemetered to both group and central control stations:

- (i) System frequency and rate of change of frequency.
- (ii) Generated active and reactive power area wise.
- (iii) Area-interconnection active and reactive power.

All the above informations are sent by telemetering to the control stations where they are displayed by suitable meters and some of these informations are fed to the computer for processing and generating commanding signals to be followed by various equipments (*e.g.*, automatic voltage regulators and other controller) for good quality of reliable and economic supply.

PROBLEMS

- 19.1. Incremental fuel costs in Rs. per megawatt hour for two units in a plant are given by

$$\frac{dF_1}{dP_1} = 0.1 P_1 + 20$$

$$\frac{dF_2}{dP_2} = 0.12 P_2 + 16$$

The minimum and maximum loads on each unit are to be 20 MW and 125 MW respectively. Determine the incremental fuel cost and the allocation of load between units for the minimum cost when loads are (i) 100 MW, (ii) 150 MW. Assume both the units are operating.

- 19.2. Determine the saving in fuel cost in Rs. per hour for the economic distribution of a total load of 200 MW between the two units of the plant described in Problem 19.1 compared with equal distribution of the same total load.
- 19.3. A system consists of two plants connected by a tie line and a load is located at plant 2. When 100 MW are transmitted from plant 1, a loss of 10 MW takes place on the tie-line. Determine the generation schedule at both the plants and the power received by the load when λ for the system is Rs. 25 per megawatt hour and the incremental fuel costs are given by the equation

$$\frac{dF_1}{dP_1} = 0.03 P_1 + 17 \text{ Rs./MWhr}$$

$$\frac{dF_2}{dP_2} = 0.06 P_2 + 19 \text{ Rs./MWhr}$$

- 19.4. If the power received by the load in the previous example is 200 MW, determine the savings in rupees per hour obtained by coordinating rather than simply including the transmission loss and not coordinating them in determining the loading of the plant.
- 19.5. The incremental fuel costs for two plants are given by

$$\frac{dF_1}{dP_1} = 0.1 P_1 + 22$$

$$\frac{dF_2}{dP_2} = 0.12 P_2 + 16$$

where F is in Rs./hr and P is in MW. If both the units operate at all times and maximum and minimum loads on each unit are 100 MW and 20 MW respectively. Determine the economic operating schedule of the plants for loads of 40 KW, 60 MW, 80 MW, 120 MW, 160 MW and 180 MW, neglecting the transmission line losses.